Problems on Linear Congruences and Reduced Residue Systems

Theorems (from Strayer)

1. **Linear congruence solvability.** The congruence $ax \equiv b \pmod{n}$ has a solution iff $d = \gcd(a, n) \mid b$. If $d \mid b$ then dividing through by d yields $(a/d)x \equiv (b/d) \pmod{n/d}$, which has a unique solution modulo n/d, and the original congruence has d distinct solutions modulo n.

2. **Multiplicative inverse.** a has an inverse modulo n iff $\gcd(a, n) = 1$.

3. **Chinese Remainder Theorem (pair).** The system $x \equiv a \pmod{m}$, $x \equiv b \pmod{n}$ has a solution iff $a \equiv b \pmod{\gcd(m, n)}$. If $\gcd(m, n) = 1$ the solution is unique modulo mn.

Definition

Reduced residue system modulo n: A set $R = \{r_1, \ldots, r_{\varphi(n)}\}$ is a reduced residue system modulo n iff

$$\gcd(r_i, n) = 1 \quad \text{for all } i, \text{ where } r_i \pmod{n} \text{ is unique residue.}$$

Problems and concise solutions

1. $84x \equiv 5 \pmod{49}$.
   $\gcd(84, 49) = 7$. Since $7 \nmid 5$, by theorem 1 there is **no solution**.

2. $12x \equiv 16 \pmod{32}$.
   $\gcd(12, 32) = 4$ and $4 \mid 16$. Divide by 4: $3x \equiv 4 \pmod{8}$. Since $\gcd(3, 8) = 1$ and $3^{-1} \equiv 3 \pmod{8}$, we get $x \equiv 3 \cdot 4 = 12 \equiv 4 \pmod{8}$. The least positive solution is $\boxed{4}$. (All solutions mod 32: $4, 12, 20, 28$.)

3. $7x \equiv 1 \pmod{24}$ (inverse of 7 mod 24).
   $\gcd(7, 24) = 1$. Extended Euclid gives $7 \cdot 7 - 2 \cdot 24 = 1$, so $7^{-1} \equiv 7 \pmod{24}$. Least positive solution: $\boxed{7}$.

4. System $x \equiv 3 \pmod{7}$, $x \equiv 4 \pmod{6}$.
   $\gcd(7, 6) = 1$. Let $x = 3 + 7k$. Then $3 + 7k \equiv 4 \pmod{6} \Rightarrow 7k \equiv 1 \pmod{6}$. Since $7 \equiv 1 \pmod{6}$, $k \equiv 1 \pmod{6}$. Take $k = 1$ gives $x = 10$. Hence $x \equiv 10 \pmod{42}$. Least positive: $\boxed{10}$.

5. Integers a, b with $\gcd(a, b) = 3$ and $a + b = 65$.
   If $\gcd(a, b) = 3$, then $3 \mid a$ and $3 \mid b$, so $3 \mid (a + b)$. But $3 \nmid 65$. Contradiction. **No such integers.**

6. Reduced residue system modulo 18.
   $\varphi(18) = 6$. Units mod 18 (least positive representatives): $\boxed{\{1, 5, 7, 11, 13, 17\}}$.

7. Is $\{-15, 66, -75, 11\}$ a reduced residue system modulo 8?
   Reduce mod 8: $-15 \equiv 1$, $66 \equiv 2$, $-75 \equiv 5$, $11 \equiv 3$. Residues $\{1, 2, 5, 3\}$. But 2 is not coprime to 8 (gcd $= 2$). Thus the set is **not** a reduced residue system modulo 8.

8. Reduced residue system modulo 7 consisting only of odd integers.
   One choice: $\boxed{\{1, 3, 5, 9, 11, 13\}}$. Modulo 7 these reduce to $\{1, 3, 5, 2, 4, 6\}$, i.e. all units.

Definitions

- A function $f : \mathbb{Z}_{>0} \to \mathbb{C}$ is *multiplicative* iff for all m, n with $\gcd(m, n) = 1$,

$$f(mn) = f(m)f(n).$$

- f is *completely multiplicative* iff for all positive integers m, n (no gcd condition),

$$f(mn) = f(m)f(n).$$

Theorems used

1. **Chinese Remainder Theorem (pair):** If $\gcd(m, n) = 1$, then

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

   Taking cardinalities gives multiplicativity of φ on coprime arguments.

2. **Wilson's theorem:** For prime p, $(p-1)! \equiv -1 \pmod{p}$.

3. **Euler's theorem:** If $\gcd(a, m) = 1$ then $a^{\varphi(m)} \equiv 1 \pmod{m}$.
   [Fermat's little theorem]: If $\gcd(a, p) = 1$ and p is prime then $a^{p-1} \equiv 1 \pmod{p}$.

4. **Multiplicativity of φ:** If $\gcd(m, n) = 1$ then $\varphi(mn) = \varphi(m)\varphi(n)$.

Problems and concise solutions

(A) φ is multiplicative but not completely multiplicative

Prime-power formula: For prime p, $\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p-1)$.
Multiplicativity: If $\gcd(m, n) = 1$ then by CRT

$$|(\mathbb{Z}/mn\mathbb{Z})^\times| = |(\mathbb{Z}/m\mathbb{Z})^\times| \cdot |(\mathbb{Z}/n\mathbb{Z})^\times|,$$

so $\varphi(mn) = \varphi(m)\varphi(n)$. (Cited: CRT.)
Not completely multiplicative: Example: $\varphi(4) = 2$ but $\varphi(2)\varphi(2) = 1 \cdot 1 = 1 \neq 2$. Thus φ is not completely multiplicative.

(B) Euclidean algorithm: $(831, 402)$

Compute:

$$\begin{aligned}
831 &= 2 \cdot 402 + 27,\\
402 &= 14 \cdot 27 + 24,\\
27 &= 1 \cdot 24 + 3,\\
24 &= 8 \cdot 3 + 0.
\end{aligned}$$

Hence $\gcd(831, 402) = 3$. Back-substitute:

$$\begin{aligned}
3 &= 27 - 1 \cdot 24\\
&= 27 - 1(402 - 14 \cdot 27) = 15 \cdot 27 - 1 \cdot 402\\
&= 15(831 - 2 \cdot 402) - 1 \cdot 402 = 15 \cdot 831 - 31 \cdot 402.
\end{aligned}$$

So $3 = 15 \cdot 831 - 31 \cdot 402$.

(C) For odd prime p: $2(p-3)! \equiv -1 \pmod p$

Using Wilson's theorem,

$$(p-1)! = (p-1)(p-2)(p-3)! \equiv (-1)(-2)(p-3)! \equiv 2(p-3)! \equiv -1 \pmod p.$$

By Wilson, $(p-1)!$, hence $2(p-3)! \equiv -1 \pmod p$.

1. If $\gcd(a,m)=1$, show $a^{\varphi(m)-1}$ is the inverse of a modulo m

   Proof: By Euler's theorem $a^{\varphi(m)} \equiv 1 \pmod m$. Multiply both sides by a^{-1} (or rearrange):

   $$a \cdot a^{\varphi(m)-1} \equiv a^{\varphi(m)} \equiv 1 \pmod m,$$

   so $a^{\varphi(m)-1}$ is an inverse of a modulo m. (Cited: Euler's theorem.)

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2. Find the least positive solution x to the congruence $x \equiv 20^{110} (\bmod 17)$.

   Proof: $x \equiv 20^{110} \pmod{17}$. Reduce base: $20 \equiv 3 \pmod{17}$. Since 17 is prime, by Fermat, $3^{16} \equiv 1 \pmod{17}$, and $110 \equiv 14 \pmod{16}$. Thus

   $$3^{110} \equiv 3^{14} = (3^8)(3^4)(3^2) \equiv 16 \cdot 13 \cdot 9 \equiv 4 \cdot 9 \equiv 36 \equiv 2 \pmod{17}.$$

   So least positive solution: $\boxed{2}$.

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3. For $x \equiv 38^{110} (\bmod 21)$.

   Proof: $x \equiv 38^{110} \pmod{21}$. Reduce base: $38 \equiv 17 \pmod{21}$. By Euler theorem and Multiplicativity of φ, $\varphi(21) = \varphi(3)\varphi(7) = 2 \cdot 6 = 12$,

   $$17^{\varphi(21)} \equiv 17^{12} \equiv 1 \pmod{21}.$$

   Thus,

   $$38^{110} \equiv 17^{110} \equiv 17^2 \equiv 289 \equiv 16 \pmod{21}.$$

   So least positive solution: $\boxed{16}$.

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Problems and concise proofs

All proofs are written concisely in LaTeX math mode and reference standard results where used.

1. Let $m > 1$ and let $\{r_1, \dots, r_{\varphi(m)}\}$ be a reduced residue system modulo m. Characterize integers j for which $\{jr_1, \dots, jr_{\varphi(m)}\}$ is also a reduced residue system modulo m.
   Answer. Exactly those j with $\gcd(j, m) = 1$.
   Proof. If $\gcd(j, m) = 1$ then multiplication by j is a bijection on $(\mathbb{Z}/m\mathbb{Z})^\times$, so the set $\{jr_i\}$ consists of distinct residues all coprime to m, hence is a reduced residue system. Conversely, if $\gcd(j, m) > 1$ then any r_i with $\gcd(r_i, m) = 1$ yields $\gcd(jr_i, m) > 1$, so the image contains nonunits and cannot be a reduced residue system.

2. If $a, b \in \mathbb{Z}$, $a \geq b > 0$, and $a = bq + r$ (with $0 \leq r < b$), show $(a, b) = (b, r)$.
   Proof. Any common divisor of a and b divides $a - bq = r$, so $\gcd(a, b) \mid \gcd(b, r)$. Conversely any common divisor of b and r divides $bq + r = a$, so $\gcd(b, r) \mid \gcd(a, b)$. Thus equality holds. (This is the Euclidean algorithm step.)

3. If $a, b \in \mathbb{Z}$ and $(a, b) = [a, b]$, what can you say about a, b?
   Answer. $a = \pm b$.
   Proof. Let $d = \gcd(a, b)$ and write $a = dx$, $b = dy$ with $\gcd(x, y) = 1$. Then $\operatorname{lcm}(a, b) = dxy$. If $d = \operatorname{lcm}(a, b)$ then $d = dxy$, so $xy = 1$, hence $x = y = \pm 1$. Thus $|a| = |b| = d$, so $a = \pm b$.

4. Let $a, b \in \mathbb{Z}$ and p prime. If $p \mid ab$ then $p \mid a$ or $p \mid b$.
   Proof. (Euclid's lemma.) If $p \nmid a$ then $\gcd(a, p) = 1$ so a has an inverse modulo p. From $ab \equiv 0 \pmod{p}$ multiply by a^{-1} to get $b \equiv 0 \pmod{p}$. Hence $p \mid b$.

5. Let p be prime and $p \nmid a$. Show there is a unique solution to $ax \equiv 1 \pmod{p}$.
   Proof. Since $\gcd(a, p) = 1$, a is a unit in $\mathbb{Z}/p\mathbb{Z}$ so an inverse x exists. If x_1, x_2 are both inverses then $a(x_1 - x_2) \equiv 0 \pmod{p}$. As $p \nmid a$, we get $p \mid (x_1 - x_2)$, so $x_1 \equiv x_2 \pmod{p}$. Thus the inverse is unique modulo p.

6. Let $a, b, c, m \in \mathbb{Z}$ with $m, c \geq 1$. If $a \equiv b \pmod{m}$ then $ac \equiv bc \pmod{mc}$.
   Proof. $a \equiv b \pmod{m}$ implies $m \mid (a - b)$, so $mc \mid c(a - b) = ac - bc$. Therefore $ac \equiv bc \pmod{mc}$.

7. Let $a, b, c, d \in \mathbb{Z}$ with $b, d \neq 0$ and $(a, b) = (c, d) = 1$. If $\dfrac{a}{b} + \dfrac{c}{d}$ is an integer, show $b = \pm d$.
   Proof. Suppose $\dfrac{a}{b} + \dfrac{c}{d} = k \in \mathbb{Z}$. Clearing denominators gives
   $$ad + bc = kbd \quad \Rightarrow \quad ad = b(kd - c).$$
   Thus $b \mid ad$. Since $\gcd(a, b) = 1$, we conclude $b \mid d$. Symmetrically $d \mid b$. Hence $|b| = |d|$, so $b = \pm d$.

Remarks: The starred items are direct instances of standard results (Euclidean algorithm step, Euclid's lemma, units mod p). The proofs above are intentionally concise and in math mode.